

MID TERM EXAMINATION

B.TECH (AI&DS, AI&ML, IIOT) — THIRD SEMESTER, NOVEMBER 2022

Paper Code: AIDS 201 / AIML 201 / IIOT 201

Subject: Data Structure

Time: 1.5 Hours

Maximum Marks: 30

Note: Q. No. 1 is compulsory. Attempt any two questions from the remaining questions. All questions carry equal marks. Only scientific calculator is allowed.

Q. No.	Question	Marks	CO	BL
Question 1 (Compulsory)				
1. (a)	Explain Space and Time Tradeoff.	2.5	1	Understand
(b)	Differentiate between Singly, Circular and Doubly Linked list.	2.5	2	Analyze
(c)	Assume that 4 bytes of storage is required to store each element of an array of integers A[12][15] . What will be the actual address of the element at A[6][5] ? Assume Base address is 1000.	2.5	1	Evaluate
(d)	Write an algorithm to search an element in an array using Linear Search.	2.5	2	Create
Question 2				
2. (a)	Using stack translate the infix expression A * (B + C) / D - E * (F + G / H) into equivalent Postfix expression.	5	1	Apply
(b)	Explain insertion and deletion process in a Queue with an example.	5	1	Understand
Question 3				
3. (a)	Write an algorithm to add a node at the end of Circular Linked List. Explain with the help of an example.	5	2	Create
(b)	Sort the following list using Quick Sort. Show the step-by-step process: 44, 25, 76, 13, 67, 60, 96, 38, 98, 89	5	2	Evaluate
Question 4				
4. (a)	Write an algorithm to reverse a singly linked list.	5	2	Understand
(b)	Write an algorithm to add an element in an array. Explain with the help of an example.	5	1	Remember

**** End of Question Paper ****